



OPTIMIZATION







Introduction

- The core problem in ML is parameter estimation (model fitting).
- We need to solve an **optimization problem**: i.e., trying to find the values for a set of variables $\theta \in \Theta$ that minimize a scalar-valued **loss function** or **cost function**: $\mathcal{L} : \Theta \rightarrow \mathbb{R}$

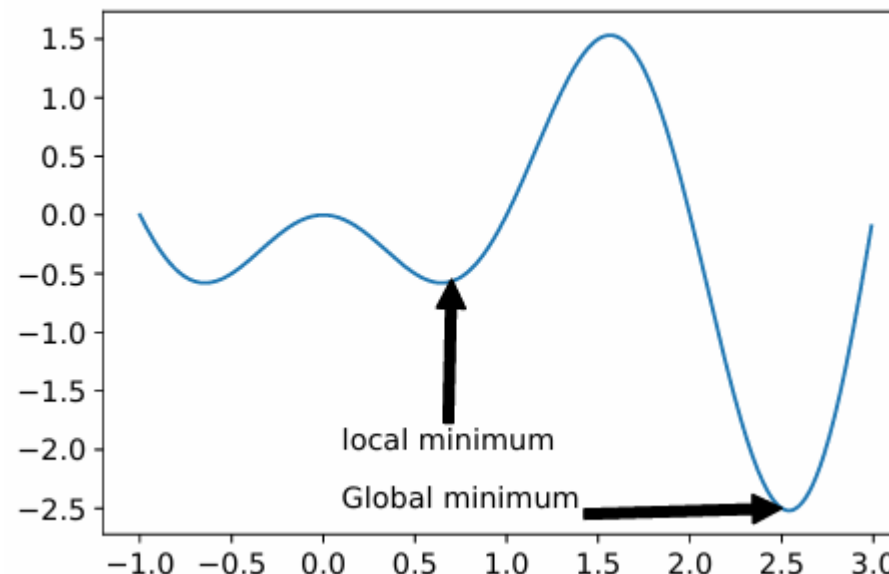
$$\theta^* = \underset{\theta \in \Theta}{\operatorname{argmin}} \mathcal{L}(\theta) \quad (1)$$

- The **parameter space** is given by $\Theta \in \mathbb{R}^D$, where D is the number of variables being optimized.
- We focus on **continuous optimization**.
- To maximize a **score function** or **reward function** $R(\theta)$, we can minimize $\mathcal{L}(\theta) = -R(\theta)$.
- The term **objective function** refers to a function we want to maximize or minimize.
- An algorithm to find an optimum of an objective function is a **solver**.

Local versus global optimization

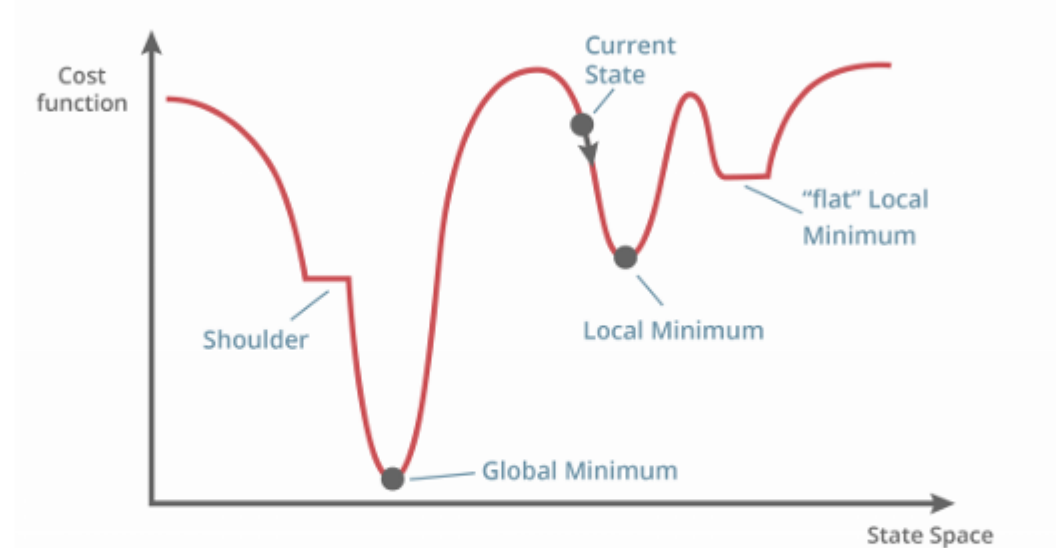
- A point that satisfies Equation 1 is called a **global optimum**. Finding such a point is called **global optimization**.
- In general, finding global optima is computationally **intractable**. We will try to find a **local optimum**.
- For continuous problem, a local optimum is a point θ^* which has lower (or equal) cost than “nearby” points.

$$\exists \delta > 0, \forall \theta \in \Theta, \text{ s.t. } \|\theta - \theta^*\| < \delta, \mathcal{L}(\theta^*) \leq \mathcal{L}(\theta) \quad (2)$$



Local versus global optimization

- A local minimum could be surrounded by other local minima with the same objective value; this is known as a **flat local minimum**.



- A point is said to be a **strict local minimum** if its cost is strictly lower than those of neighboring points.

$$\exists \delta > 0, \forall \theta \in \Theta, \theta \neq \theta^* : \|\theta - \theta^*\| < \delta, \mathcal{L}(\theta^*) < \mathcal{L}(\theta) \quad (3)$$

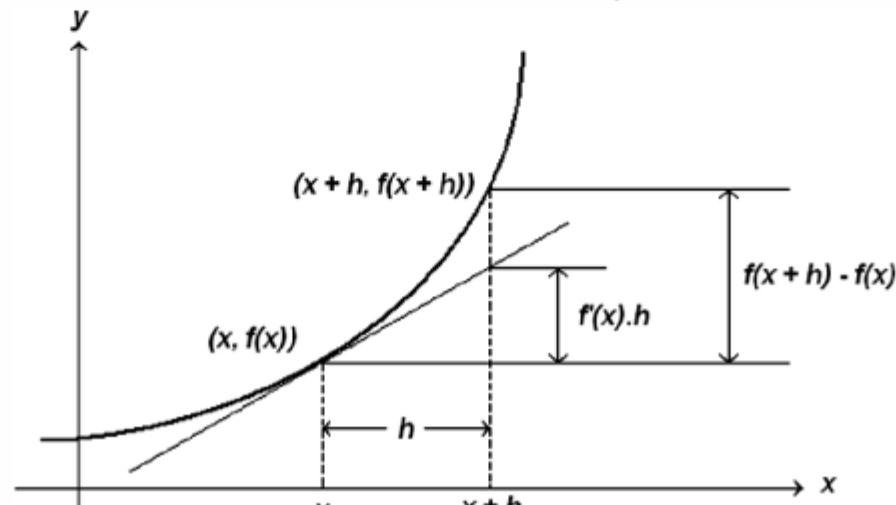
Derivatives

- The topic of **calculus** concerns computing “**rates of change**” of functions as we vary their inputs.
- Consider a scalar-argument function $f : \mathbb{R} \rightarrow \mathbb{R}$. Its **derivative** at a point a is the quantity

$$f'(x) \triangleq \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

assuming the limit exists.

- This measures how quickly the output changes when we move a small distance in the input space away from x (i.e., the “rate of change”).





Derivatives

- We can compute a **finite difference** approximation to the derivative by using a finite step size h

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \underbrace{\frac{f(x+h) - f(x)}{h}}_{\text{forward difference}} \\ &= \lim_{h \rightarrow 0} \underbrace{\frac{f(x+h/2) - f(x-h/2)}{h}}_{\text{central difference}} \\ &= \lim_{h \rightarrow 0} \underbrace{\frac{f(x) - f(x-h)}{h}}_{\text{backward difference}} \end{aligned}$$

- The smaller the step size h , the better the estimate.



Gradients

- We extend the notion of derivatives to handle vector-argument functions, $f : \mathbb{R}^n \rightarrow \mathbb{R}$, by defining the **partial derivative** of f with respect to x_i to be

$$\frac{\partial f}{\partial x_i} = \lim_{h \rightarrow 0} \frac{f(\mathbf{x} + h\mathbf{e}_i) - f(\mathbf{x})}{h}$$

where $\mathbf{e}_i$ is the i 'th unit vector, $\mathbf{e}_i = (0, \dots, 1, \dots, 0)$ with the i 'th element = 1 and all the other elements are 0.

- The **gradient** of f at a point $\mathbf{x}$ is the vector of its partial derivatives

$$\mathbf{g} = \frac{\partial f}{\partial \mathbf{x}} = \nabla f = \begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{pmatrix} = \frac{\partial f}{\partial x_1} \mathbf{e}_1 + \dots + \frac{\partial f}{\partial x_n} \mathbf{e}_n$$

- To emphasize the point at which the gradient is evaluated, we write

$$\mathbf{g}(\mathbf{x}^*) \triangleq \left. \frac{\partial f}{\partial \mathbf{x}} \right|_{\mathbf{x}^*}$$



Directional derivative

- The **directional derivative** measures how much the function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ changes along a direction $\mathbf{v}$ in space.

$$D_{\mathbf{v}}f(\mathbf{x}) = \lim_{h \rightarrow 0} \frac{f(\mathbf{x} + h\mathbf{v}) - f(\mathbf{x})}{h}$$

- We can approximate this numerically using 2 function calls to f , regardless of n .
- By contrast, a numerical approximation to the standard gradient vector takes $n + 1$ calls (or $2n$ if using central differences).
- The directional derivative along $\mathbf{v}$ is the scalar product of the gradient $\mathbf{g}$ and the vector $\mathbf{v}$:

$$D_{\mathbf{v}}f(\mathbf{x}) = \nabla f(\mathbf{x}) \cdot \mathbf{v}$$



Directional derivative

Example: Let $f(x, y) = x^2y$. Find the derivative of f in the direction $(1, 2)$ at the point $(3, 2)$.



Directional derivative

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- The gradient $\nabla f(x, y)$ is:

$$\nabla f(x, y) = \begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{pmatrix} = \begin{pmatrix} 2xy \\ x^2 \end{pmatrix}$$

$$\nabla f(3, 2) = \begin{pmatrix} 12 \\ 9 \end{pmatrix} = 12 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 9 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 12\mathbf{e}_1 + 9\mathbf{e}_2$$

- Let $\mathbf{u} = u_1\mathbf{e}_1 + u_2\mathbf{e}_2$ be a unit vector. The derivative of f in the direction of $\mathbf{u}$ at $(3, 2)$ is:

$$\begin{aligned} D_{\mathbf{u}}f(3, 2) &= \nabla f(3, 2) \cdot \mathbf{u} \\ &= (12\mathbf{e}_1 + 9\mathbf{e}_2) \cdot (u_1\mathbf{e}_1 + u_2\mathbf{e}_2) \\ &= 12u_1 + 9u_2 \end{aligned}$$



Directional derivative

- The unit vector in the direction of vector $(1,2)$ is:

$$\mathbf{u} = \frac{(1, 2)}{\|(1, 2)\|} = \frac{(1, 2)}{\sqrt{1^2 + 2^2}} = \frac{(1, 2)}{\sqrt{5}} = (1/\sqrt{5}, 2/\sqrt{5})$$

- The directional derivative at $(3,2)$ in the direction of $(1,2)$ is:

$$\begin{aligned} D_{\mathbf{u}}f(3, 2) &= 12u_1 + 9u_2 \\ &= \frac{12}{\sqrt{5}} + \frac{18}{\sqrt{5}} = \frac{30}{\sqrt{5}} \end{aligned}$$

- We normalize vector $(1,2)$ so that the directional derivative is independent of its magnitude and depending only on its direction.



Directional derivative

Example 2: Let $f(x, y) = x^2y$. Find the derivative of f in the direction of $(2, 1)$ at the point $(3, 2)$.



Directional derivative

Example 2: Let $f(x, y) = x^2y$. Find the derivative of f in the direction of $(2,1)$ at the point $(3,2)$.

- The unit vector in the direction of $(2,1)$ is:

$$\mathbf{u} = \frac{(2, 1)}{\sqrt{5}} = (2/\sqrt{5}, 1/\sqrt{5})$$

- The directional derivative of f at $(3,2)$ in the direction of $(2,1)$ is:

$$\begin{aligned} D_{\mathbf{u}}f(3, 2) &= 12u_1 + 9u_2 \\ &= \frac{24}{\sqrt{5}} + \frac{9}{\sqrt{5}} = \frac{33}{\sqrt{5}} \end{aligned}$$



Directional derivative

Questions:

- At a point $\mathbf{a}$, in which direction $\mathbf{u}$ is the directional derivative $D_{\mathbf{u}}f(\mathbf{a})$ maximal?
- What is the directional derivative in that direction $D_{\mathbf{u}}f(\mathbf{a}) = ?$

The relationship between the **gradient** and the **directional derivative**:

$$\begin{aligned} D_{\mathbf{u}}f(\mathbf{a}) &= \nabla f(\mathbf{a}) \cdot \mathbf{u} \\ &= \|\nabla f(\mathbf{a})\| \|\mathbf{u}\| \cos \theta \quad [\theta \text{ is the angle between } \mathbf{u} \text{ and the gradient.}] \\ &= \|\nabla f(\mathbf{a})\| \cos \theta \quad [\mathbf{u} \text{ is a unit vector.}] \end{aligned}$$

The maximal value of $D_{\mathbf{u}}f(\mathbf{a})$ occurs when $\mathbf{u}$ and $\nabla f(\mathbf{a})$ point in the same direction (i.e., $\theta = 0$).

Jacobian

- Consider a function that maps a vector to another vector, $\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^m$. The **Jacobian matrix** of this function is an $m \times n$ matrix of partial derivatives:

$$\mathbf{J}_{\mathbf{f}}(\mathbf{x}) = \frac{\partial \mathbf{f}}{\partial \mathbf{x}^{\top}} \triangleq \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{pmatrix} = \begin{pmatrix} \nabla f_1(\mathbf{x})^{\top} \\ \vdots \\ \nabla f_m(\mathbf{x})^{\top} \end{pmatrix}$$



Hessian

- For a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ that is twice differentiable, the **Hessian matrix** is the (symmetric) $n \times n$ matrix of second partial derivatives

$$\mathbf{H}_f = \frac{\partial^2 f}{\partial \mathbf{x}^2} = \nabla^2 f = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{pmatrix}$$

- The Hessian is the Jacobian of the gradient.



Geometric meaning

- If we follow the direction $\mathbf{d}$ from $\mathbf{x}$, we can define a uni-dimensional function $g(\alpha)$:

$$g(\alpha) = f(\mathbf{x} + \alpha \mathbf{d})$$

$$g'(\alpha) = \mathbf{d}^\top \nabla f(\mathbf{x} + \alpha \mathbf{d})$$

$$g''(\alpha) = \mathbf{d}^\top \nabla^2 f(\mathbf{x} + \alpha \mathbf{d}) \mathbf{d}$$

- Interpretation

$$g'(0) = \mathbf{d}^\top \nabla f(\mathbf{x}) \quad [\text{directional derivative}]$$

$$g''(0) = \mathbf{d}^\top \nabla^2 f(\mathbf{x}) \mathbf{d} \quad [\text{directional curvature}]$$

- If $g''(0)$ is non-negative with a certain $\mathbf{d}$: f is convex in direction $\mathbf{d}$.
- If $g''(0)$ is non-negative for all $\mathbf{d}$: $\nabla^2 f(\mathbf{x})$ is positive semidefinite $\rightarrow f$ is convex at $\mathbf{x}$.

Optimality conditions for local vs global optima



- For continuous, twice differentiable functions, we can characterize the **points** which correspond to **local optima**.
- Let $g(\theta) = \nabla \mathcal{L}(\theta)$ be the **gradient** vector, and $H(\theta) = \nabla^2 \mathcal{L}(\theta)$ be the **Hessian** matrix.
- Consider a point $\theta^* \in \mathbb{R}^D$, and let $g^* = g(\theta)|_{\theta^*}$ be the gradient at that point, and $H^* = H(\theta)|_{\theta^*}$ be the corresponding Hessian.
 - 1 **Necessary conditions:** If θ^* is a local minimum, then we must have $g^* = 0$ (i.e., θ^* must be a **stationary point**), and H^* must be positive semi-definite.
 - Suppose we were at a point θ^* at which the gradient is non-zero.
 - At such a point, we could decrease the function by following the negative gradient a small distance, so this would not be optimal.
 - So the gradient must be zero.
 - 2 **Sufficient conditions:** If $g^* = 0$ and H^* is positive definite, then θ^* is a local optimum.
 - Why a zero gradient is not sufficient?
 - The stationary point could be a local minimum, local maximum, or **saddle** point.



Global optimizers

- We classify a stationary point of a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ as a **global minimizer** if the Hessian matrix of f is positive semidefinite **everywhere**,
- and as a **global maximizer** if the Hessian matrix is negative semidefinite everywhere.
- If the Hessian matrix is positive definite, or negative definite, the minimizer and maximizer (respectively) is strict.

$$\text{Let } f(x_1, x_2) = (x_1^2 + x_2^2 - 1)^2 + (x_2^2 - 1)^2$$

Example:

$$f(x, y, z) = x^2 + y^2 + z^2 - xy + yz - xz.$$

Find the extreme point of f

Solution:

$$\nabla f(x, y, z) = (2x - y - z, 2y - x + z, 2z + y - x).$$

$$\text{Solve: } \nabla f(x, y, z) = 0 \Leftrightarrow X^*(x, y, z) = (0, 0, 0)$$

$$\nabla^2 f(x, y, z) = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & 1 \\ -1 & 1 & 2 \end{bmatrix}$$

$$\Delta_1 = |2| = 2 > 0; \Delta_2 = \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} = 3 > 0; \Delta_3 = \begin{vmatrix} 2 & -1 & -1 \\ -1 & 2 & 1 \\ -1 & 1 & 2 \end{vmatrix} = 4 > 0$$

All the principal minors are positive $\Rightarrow$ Hessian matrix is positive definite

X^* is a relative minimum point

Example:

$$f(x, y, z) = e^{x-y} + e^{y-x} + e^{x^2} + z^2.$$

Find the extreme point of f

Solution:

$$\nabla f(x, y, z) = (e^{x-y} - e^{y-x} + 2xe^{x^2}, -e^{x-y} + e^{y-x}, 2z)$$

$$\begin{aligned} \text{Solve: } \nabla f(x, y, z) = 0 &\Leftrightarrow z=0; x=y; x=0 \\ &\Leftrightarrow (x, y, z) = (0, 0, 0) \end{aligned}$$

$$\nabla^2 f(x, y, z) = \begin{bmatrix} e^{x-y} + e^{y-x} + 4x^2e^{x^2} + 2e^{x^2} & -e^{x-y} - e^{y-x} & 0 \\ -e^{x-y} - e^{y-x} & e^{x-y} + e^{y-x} & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$\nabla^2 f(0, 0, 0) = \begin{pmatrix} 4 & -2 & 0 \\ -2 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} = A$$

$$|A - \lambda I_3| = \begin{vmatrix} 4-\lambda & -2 & 0 \\ -2 & 2-\lambda & 0 \\ 0 & 0 & 2-\lambda \end{vmatrix} = (2-\lambda)(\lambda^2 - 6\lambda + 4)$$

The eigenvalues of the matrix A are:
 $\lambda=2; \lambda=3+\sqrt{5}; \lambda=3-\sqrt{5}$

Because all eigenvalues are positive
 $\Rightarrow$ Hessian matrix is positive definite

Therefore, the extreme point (or critical point) $(0, 0, 0)$ is a relative minimum point (strict global minimizer)



Global optimizers

Let $f(x_1, x_2) = (x_1^2 + x_2^2 - 1)^2 + (x_2^2 - 1)^2$

- The gradient is $\nabla f(\mathbf{x}) = 4 \begin{pmatrix} (x_1^2 + x_2^2 - 1)x_1 \\ (x_1^2 + x_2^2 - 1)x_2 + (x_2^2 - 1)x_2 \end{pmatrix}$
- The stationary points are $(0,0)$, $(1,0)$, $(-1,0)$, $(0,1)$, $(0,-1)$.
- The Hessian is $\nabla^2 f(\mathbf{x}) = 4 \begin{pmatrix} 3x_1^2 + x_2^2 - 1 & 2x_1x_2 \\ 2x_1x_2 & x_1^2 + 6x_2^2 - 2 \end{pmatrix}$
- Since $\nabla^2 f(0,0) = 4 \begin{pmatrix} -1 & 0 \\ 0 & -2 \end{pmatrix} \prec 0$, it follows that $(0,0)$ is a **strict local maximum** point.



Global optimizers

- $\nabla^2 f(1, 0) = \nabla^2 f(-1, 0) = 4 \begin{pmatrix} 2 & 0 \\ 0 & -1 \end{pmatrix}$, which is an indefinite matrix.
- Hence $(1, 0)$ and $(-1, 0)$ are **saddle points**.
- $\nabla^2 f(0, 1) = \nabla^2 f(0, -1) = 4 \begin{pmatrix} 0 & 0 \\ 0 & 4 \end{pmatrix}$, which is positive semidefinite.
- The fact that the Hessian matrices of f at $(0, 1)$ and $(0, -1)$ are positive semidefinite is **not enough** to conclude that these are local minimum points; they **might be** saddle points.
- However, in this case, since $f(0, 1) = f(0, -1) = 0$ and the function is lower bounded by zero, $(0, 1)$ and $(0, -1)$ are **global minimum points**.
- Because there are two global minimum points, they are **nonstrict** global minima, but they are **strict** local minimum points, since each has a neighborhood in which it is the unique minimizer.



Constrained vs unconstrained optimization

- In **unconstrained optimization**, we find any value in the parameter space Θ that minimizes the loss.
- We can also have a set of **constraints** $\mathcal{C}$ on the allowable values.
- We partition the set of constraints $\mathcal{C}$ into:
 - **Inequality constraints**: $g_j(\theta) \leq 0$ for $j \in \mathcal{I}$.
 - **Equality constraints**: $h_k(\theta) = 0$ for $k \in \mathcal{E}$.
- The **feasible set** is the subset of the parameter space that satisfies the constraints:

$$\mathcal{C} = \{\theta : g_j(\theta) \leq 0 : j \in \mathcal{I}, h_k(\theta) = 0 : k \in \mathcal{E}\} \subseteq \mathbb{R}^D$$

- Our **constrained optimization** problem is

$$\hat{\theta}^* = \underset{\theta \in \mathcal{C}}{\operatorname{argmin}} \mathcal{L}(\theta)$$

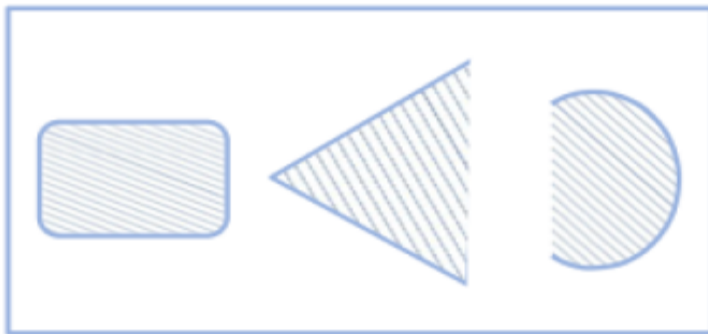
- If $\mathcal{C} = \mathbb{R}^D$, it is called **unconstrained optimization**.

Convex sets

- In **convex optimization**, the objective is a convex function defined over a convex set.
- In such problems, every local minimum is also a global minimum.
- Many models are designed so that their training objectives are convex.
- We say $\mathcal{S}$ is a **convex set** if, for any $x, x' \in \mathcal{S}$, we have

$$\lambda x + (1 - \lambda)x' \in \mathcal{S}, \forall \lambda \in [0, 1]$$

- If we draw a line from x to x' , all points on the line lie inside the set.



Convex

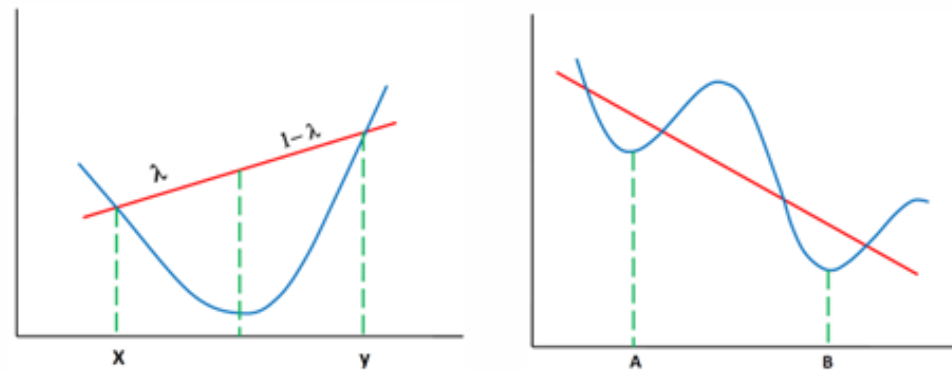


Not Convex

Convex functions

- $f(x)$ is called a convex function if it is defined on a convex set, and if, for any $x, y \in \mathcal{S}$, and for any $0 \leq \lambda \leq 1$, we have:

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$$



- A function is **strictly convex** if the inequality is strict.
- A function is **concave** if $-f(x)$ is convex.
- A function can be neither convex nor concave.
- Some examples of 1d convex functions: x^2 , e^{ax} , $-\log(x)$, $x^a (a > 1, x > 0)$, $|x|^a (a \geq 1)$, $x \log x (x > 0)$.



Convex functions

Theorem

Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is twice differentiable over its domain. Then f is convex iff $\mathbf{H} = \nabla^2 f(\mathbf{x})$ is positive semi-definite for all $\mathbf{x} \in \text{dom}(f)$. Furthermore, f is strictly convex if $\mathbf{H}$ is positive definite.

- For example, consider the quadratic form

$$f(\mathbf{x}) = \mathbf{x}^\top \mathbf{A} \mathbf{x}$$



Convex functions

Theorem

Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is twice differentiable over its domain. Then f is convex iff $\mathbf{H} = \nabla^2 f(\mathbf{x})$ is positive semi-definite for all $\mathbf{x} \in \text{dom}(f)$. Furthermore, f is strictly convex if $\mathbf{H}$ is positive definite.

- For example, consider the quadratic form

$$f(\mathbf{x}) = \mathbf{x}^\top \mathbf{A} \mathbf{x}$$

- This is convex if $\mathbf{A}$ is positive semi-definite.
- This is strictly convex if $\mathbf{A}$ is positive definite.
- It is neither convex nor concave if $\mathbf{A}$ has eigenvalues of mixed sign.
- Intuitively, a convex function is shaped like a bowl.

Quadratic form

Let $f(x) = x^T Ax + bx + c$ where A is symmetric.

Then f is convex if and only if A is positive semidefinite, independently of b, c .

$$f(x_1, x_2, x_3) = 3x_1^2 + 3x_2^2 + 4x_3^2 + 4x_1x_2 + 2x_1x_3 + 2x_2x_3 - x_1 - 2x_2 - 3x_3$$

- 1) Determine whether the following function is convex or concave.
- 2) Find the global minimum/maximum of the given f

Solution:

$$f(x_1, x_2, x_3) = x^T Ax + b^T x \quad A = \begin{pmatrix} 3 & 2 & 1 \\ 2 & 3 & 1 \\ 1 & 1 & 4 \end{pmatrix}; \quad b = \begin{pmatrix} -1 \\ -2 \\ -3 \end{pmatrix};$$

$$\text{then: } \nabla f(x_1, x_2, x_3) = 2Ax + b, \quad \nabla^2 f(x_1, x_2, x_3) = 2A.$$

$$\nabla f(x_1, x_2, x_3) = (6x_1 + 4x_2 + 2x_3 - 1, 4x_1 + 6x_2 + 2x_3 - 2, 2x_1 + 2x_2 + 8x_3 - 3)$$

Matrix calculus- Derivative Functions

y	$\frac{\partial y}{\partial \mathbf{x}}$
$\mathbf{Ax}$	$\mathbf{A}$
$\mathbf{x}^T \mathbf{A}$	$\mathbf{A}^T$
$\mathbf{x}^T \mathbf{x}$	$2\mathbf{x}^T$
$\mathbf{x}^T \mathbf{Ax}$	$\mathbf{x}^T \mathbf{A} + \mathbf{x}^T \mathbf{A}^T$

- $|A - \lambda I| = \begin{vmatrix} 3 - \lambda & 2 & 1 \\ 2 & 3 - \lambda & 1 \\ 1 & 1 & 4 - \lambda \end{vmatrix} = (\lambda - 1)(-\lambda^2 + 9\lambda - 18)$

$$|A - \lambda I| = 0 \Leftrightarrow (\lambda - 1)(-\lambda^2 + 9\lambda - 18) = 0 \Leftrightarrow \lambda = 1; \lambda = 3; \lambda = 6$$

Because of all the eigenvalues are positive, therefore A is positive definite.

$\Rightarrow f(\mathbf{X})$ is strictly convex function $\Rightarrow f$ has a unique global minimum,

$$\nabla f(x_1, x_2, x_3) = 2Ax + b$$

$$\nabla f(x_1, x_2, x_3) = (6x_1 + 4x_2 + 2x_3 - 1, 4x_1 + 6x_2 + 2x_3 - 2, 2x_1 + 2x_2 + 8x_3 - 3)$$

The global minimum point is the solution of equation system :

$$\nabla f(x_1, x_2, x_3) = 0 \Leftrightarrow x_1 = -\frac{1}{6}, x_2 = \frac{1}{3}, x_3 = \frac{1}{3}$$

$$\text{Min } f = f\left(-\frac{1}{6}, \frac{1}{3}, \frac{1}{3}\right)$$

Example

Determine whether the following function is convex or concave.

$$f(x_1, x_2) = 2x_1^3 - 6x_2^2$$

Solution:

$$H(\mathbf{X}) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_1 \partial x_2} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix} = \begin{bmatrix} 12x_1 & 0 \\ 0 & -12 \end{bmatrix}$$

Here

$$\begin{aligned} \frac{\partial^2 f}{\partial x_1^2} &= 12x_1 \leq 0 \text{ for } x_1 \leq 0 \\ &\geq 0 \text{ for } x_1 \geq 0 \end{aligned}$$

$$\begin{aligned} H_{22} &= -12 \leq 0 \text{ for } x_1 \leq 0 \\ &\geq 0 \text{ for } x_1 \geq 0 \end{aligned}$$

Hence $H(\mathbf{X})$ will be semidefinite negative and $f(\mathbf{X})$ is concave for $x_1 \leq 0$

Example

- 1) Determine whether the following function is convex or concave.
- 2) Find the global minimum/maximum of the given f

$$f(x_1, x_2, x_3) = 4x_1^2 + 3x_2^2 + 5x_3^2 + 6x_1x_2 + x_1x_3 - 3x_1 - 2x_2 + 15$$

Example

Solution:

$$f(x_1, x_2, x_3) = 4x_1^2 + 3x_2^2 + 5x_3^2 + 6x_1x_2 + x_1x_3 - 3x_1 - 2x_2 + 15$$

$$\nabla f(x_1, x_2, x_3) = (8x_1 + 6x_2 + x_3 - 3, 6x_2 + 6x_1 - 2, 10x_3 + x_1 + x_3)$$

$$\nabla^2 f(x_1, x_2, x_3) = H(\mathbf{X}) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \frac{\partial^2 f}{\partial x_1 \partial x_3} \\ \frac{\partial^2 f}{\partial x_1 \partial x_2} & \frac{\partial^2 f}{\partial x_2^2} & \frac{\partial^2 f}{\partial x_2 \partial x_3} \\ \frac{\partial^2 f}{\partial x_1 \partial x_3} & \frac{\partial^2 f}{\partial x_2 \partial x_3} & \frac{\partial^2 f}{\partial x_3^2} \end{bmatrix} = \begin{bmatrix} 8 & 6 & 1 \\ 6 & 6 & 0 \\ 1 & 0 & 10 \end{bmatrix}$$

Example

Solution cont'd:

Here the principal minors are given by:

$$|8| = 8 > 0$$

$$\begin{vmatrix} 8 & 6 \\ 6 & 6 \end{vmatrix} = 12 > 0$$

$$\begin{vmatrix} 8 & 6 & 1 \\ 6 & 6 & 0 \\ 1 & 0 & 10 \end{vmatrix} = 114 > 0$$

and hence the matrix $\mathbf{H}(\mathbf{X})$ is positive definite for all real values of x_1, x_2, x_3 . Therefore $f(\mathbf{X})$ is strictly convex function.

2) Find the global minimum/maximum:

$$\nabla f(x_1, x_2, x_3) = 0 \Leftrightarrow x_1 = \frac{10}{9}, x_2 = -\frac{11}{57}, x_3 = -\frac{1}{9}$$

$f(\mathbf{X})$ is strictly convex function then f has a unique global minimum, and $\min f = f(\frac{10}{9}, -\frac{11}{57}, -\frac{1}{9}) = 15,75$



First-order methods

- We consider **iterative** optimization methods that leverage **first order** derivatives of the objective function.
- They compute which directions point “downhill”, but ignore curvature information.
- All these algorithms require the user specify a starting point θ_0 .
- At each iteration t , an update is performed

$$\theta_{t+1} = \theta_t + \rho_t d_t$$

where ρ_t is the **step size** or **learning rate**, and d_t is a **descent direction**, e.g, the negative of the **gradient** given by $g_t = \nabla_{\theta} \mathcal{L}(\theta)|_{\theta_t}$.

- The update steps are continued until a **stationary point** is reached, where the gradient is zero.



First-order methods- Descent direction

- A direction d is a **descent direction** if there is a small enough (but nonzero) amount ρ that we can move in direction d and be guaranteed to decrease the function value.
- We require there exists an $\rho_{max} > 0$ such that

$$\mathcal{L}(\theta + \rho d) < \mathcal{L}(\theta)$$

for all $0 < \rho < \rho_{max}$.

- The gradient at the current iterate,

$$g_t \triangleq \nabla \mathcal{L}(\theta)|_{\theta_t} = \nabla \mathcal{L}(\theta_t) = g(\theta_t)$$

points in the direction of maximal increase in f , so the negative gradient is a descent direction.



First-order methods- Descent direction

- Any direction $\mathbf{d}$ is also a descent direction if the angle θ between $\mathbf{d}$ and $-\mathbf{g}_t$ is less than 90 degrees and satisfies

$$\mathbf{d}^\top \mathbf{g}_t = \|\mathbf{d}\| \|\mathbf{g}_t\| \cos(\theta) < 0$$

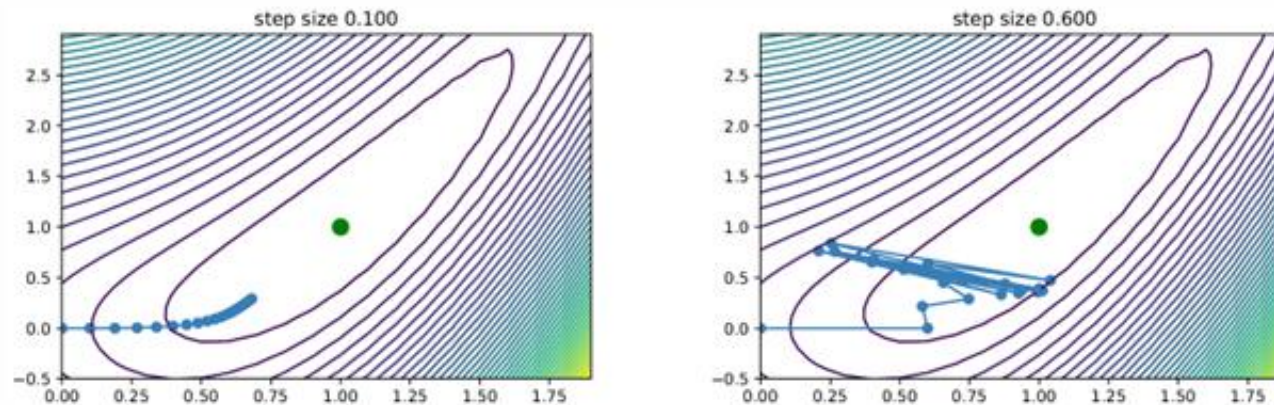
- The best choice would be to pick $\mathbf{d}_t = -\mathbf{g}_t$.
- This is the direction of **steepest descent**.

Step size (learning rate)

- The sequence of step sizes $\{\rho_t\}$ is called the **learning rate schedule**.
- The simplest method is to use constant step size, $\rho_t = \rho$.
- However, if it is too large, the method may fail to converge. If it is too small, the function will converge but very slowly.
- Example:

$$\mathcal{L}(\boldsymbol{\theta}) = 0.5(\theta_1^2 - \theta_2)^2 + 0.5(\theta_1 - 1)^2$$

- Pick our descent direction $\mathbf{d}_t = -\mathbf{g}_t$. Consider $\rho_t = 0.1$ vs $\rho_t = 0.6$:





Step size (learning rate)-Line search

- The **optimal step size** can be found by finding the value that maximally decreases the objective along the chosen direction by solving the 1d minimization problem

$$\rho_t = \operatorname{argmin}_{\rho > 0} \phi_t(\rho) = \operatorname{argmin}_{\rho > 0} \mathcal{L}(\boldsymbol{\theta}_t + \rho \mathbf{d}_t)$$

- This is **line search**: we are searching along the line defined by $\mathbf{d}_t$.
- $\phi_t(\rho) = \mathcal{L}(\boldsymbol{\theta}_t + \rho \mathbf{d}_t)$ is a convex function of an affine function of ρ , for fixed $\boldsymbol{\theta}_t$ and $\mathbf{d}$.
- If the loss is convex, this subproblem is also convex.



Step size (learning rate)-Line search

- Example, consider the quadratic loss

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{2}\boldsymbol{\theta}^\top \mathbf{A}\boldsymbol{\theta} + \mathbf{b}^\top \boldsymbol{\theta} + c$$

- Compute the derivatives of $\phi(\rho) = \mathcal{L}(\boldsymbol{\theta} + \rho \mathbf{d})$ gives

$$\begin{aligned}\frac{d\phi(\rho)}{d\rho} &= \frac{d}{d\rho} \left[\frac{1}{2}(\boldsymbol{\theta} + \rho \mathbf{d})^\top \mathbf{A}(\boldsymbol{\theta} + \rho \mathbf{d}) + \mathbf{b}^\top (\boldsymbol{\theta} + \rho \mathbf{d}) + c \right] \\ &= \mathbf{d}^\top \mathbf{A}(\boldsymbol{\theta} + \rho \mathbf{d}) + \mathbf{d}^\top \mathbf{b} \\ &= \mathbf{d}^\top (\mathbf{A}\boldsymbol{\theta} + \mathbf{b}) + \rho \mathbf{d}^\top \mathbf{A} \mathbf{d}\end{aligned}$$

- Solving for $\frac{d\phi(\rho)}{d\rho} = 0$ gives

$$\rho = -\frac{\mathbf{d}^\top (\mathbf{A}\boldsymbol{\theta} + \mathbf{b})}{\mathbf{d}^\top \mathbf{A} \mathbf{d}}$$

- This is **exact line search**. There are several methods, such as **Armijo backtracking method**, that try to ensure reduction in the objective function without spending too much time trying to solve